IV Infusions Practice 2

Module 3 - Intravenous Infusions · Daily practice sheet · No date

Name _____ Date _____

An antibiotic drug is to be given to an adult male patient (162 lb, 58 years old) by IV infusion. The drug is supplied in sterile vials containing 30 mL of the antibiotic solution at a concentration of 125 mg/mL. Assume the drug follows the pharmacokinetics of a one-compartment open model. The apparent volume of distribution of this drug is 0.5 L/kg, and the elimination half-life is 3.1 hours.

a What rate in milliliters per hour would you infuse this this drug to obtain a steady-state concentration of 20 mcg/mL in this patient?

b What loading dose would you suggest?

c How many milliliters of the antibiotic solution are required for the loading dose?

d What is the expected concentration of drug in the plasma at the end of a 6 hour infusion without a loading dose?

e What is the concentration of drug in the plasma 12 hours after the cessation of the 6-hour infusion (no loading dose)?

f If an appropriate loading dose were administered with the simultaneous IV infusion, how much time after the cessation of the infusion would be required for the concentration to reach 5 mcg/mL?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

a

Her key

Answer: 1.32 mL/hr

WHAT THE STEM GIVES

weight	162 lb	$\frac{162}{2.2} = 73.64$ kg
age	58 years old	*not needed*
vial	30 mL	*not needed for the rate*
concentration	125 mg/mL	converts mg/hr to mL/hr
V_D	0.5 L/kg	times 73.64 kg gives 36.82 L
t½	3.1 hours	$k = \frac{0.693}{3.1} = 0.2235\ hr^{-1}$
C_{ss}	20 mcg/mL	the target, 20 mg/L

THE WORKING, ONE LINE AT A TIME

UNIT

$$W = \frac{162\ lb}{2.2\ lb/kg} = 73.64\ kg;\ C_{ss} = 20\ mcg/mL = 20\ mg/L$$

Pounds over lb/kg leaves kilograms. 1 mcg/mL is 1 mg/L, so the target is 20 mg/L.

SETUP

$$R = C_{ss} \times Cl_T = C_{ss} \times V_D \times k$$

At steady state the rate in equals the rate out, so the rate that holds 20 mg/L is the target times clearance, $Cl_T = V_D \times k$.

ALGEBRA

$$R = (20\ mg/L)(0.5\ L/kg)(73.64\ kg)(\frac{0.693}{3.1\ hr}) = 164.6\ mg/hr$$

L/kg × kg = L and L × hr^{-1} = L/hr; mg/L × L/hr leaves mg/hr. The clearance is 8.23 L/hr.

UNIT

$$R = (164.6\ mg/hr)(\frac{1\ mL}{125\ mg}) = 1.32\ mL/hr$$

Each mL of solution holds 125 mg, so dividing by 125 mg/mL cancels the mg and leaves mL/hr: 1.317 mL/hr, reported as 1.32.

> **DOES THE NUMBER MAKE SENSE?**
> Clearance is about 8 L/hr, so 164.6 mg must enter each hour to hold 20 mg/L; at 125 mg in each mL that is a little over 1 mL/hr, and 1.32 mL/hr fits.

b

Her key

Answer: 738 mg

WHAT THE STEM GIVES

weight	162 lb	$\frac{162}{2.2}$ = 73.64 kg, for $C_{ss} \times V_D$
vial	30 mL	*not needed*
concentration	125 mg/mL	*not needed until part c*
V_D	0.5 L/kg	$C_{ss} \times V_D$ gives the same dose
t½	3.1 hours	k = 0.2235 hr^{-1}, the denominator
R	165 mg/hr	the numerator
age	58 years old	*not needed*
R	1.32 mL/hr	the same rate in mL/hr; not needed for D_L
C_{ss}	20 mcg/mL	$C_{ss} \times V_D$ gives 736 mg, the dose before her rounding

THE WORKING, ONE LINE AT A TIME

SETUP

$$D_L = \frac{R}{k}, \text{ with } k = \frac{0.693}{3.1 \text{ hr}} = 0.2235 \text{ hr}^{-1}$$

A loading dose puts the steady-state amount in the body at once; that amount is the infusion rate over the elimination rate constant.

ALGEBRA

$$D_L = \frac{165 \text{ mg/hr}}{0.2235 \text{ hr}^{-1}} = 738 \text{ mg}$$

mg/hr over hr^{-1}: the hours cancel and mg is left. Her key rounds the rate from part a to 165 mg/hr first.

ALGEBRA

Check: $D_L = C_{ss} \times V_D = (20 \text{ mg/L})(0.5 \text{ L/kg})(73.64 \text{ kg}) = 736 \text{ mg}$

mg/L × L leaves mg. The 2 mg difference comes only from rounding the rate to 165 mg/hr.

> **DOES THE NUMBER MAKE SENSE?**
> The loading dose is the amount the body holds at the target, 20 mg/L times 0.5 L/kg times 73.64 kg, 736 mg; 738 mg from the rounded rate fits.

$C_{ss} \times V_D$ gives 736 mg; her key divides the rate rounded to 165 mg/hr by k and gets 738 mg. Both are accepted.

c

Her key

Answer: 5.9 mL

WHAT THE STEM GIVES

concentration	125 mg/mL	the denominator
vial	30 mL	one vial holds more than enough
D_L	738 mg	the amount to draw up
weight	162 lb	*not needed: D_L is given*
age	58 years old	*not needed*
V_D	0.5 L/kg	*not needed: D_L is given*

t½	3.1 hours	*not needed: D_L is given*

THE WORKING, ONE LINE AT A TIME

SETUP

$$\text{Volume} = \frac{\text{amount}}{\text{concentration}}$$

Concentration is amount over volume, so the volume of solution that holds the loading dose is the dose over the concentration of the vial.

ALGEBRA

$$\text{Volume} = (738\ \text{mg})\left(\frac{1\ \text{mL}}{125\ \text{mg}}\right) = 5.9\ \text{mL}$$

mg × mL/mg: the mg cancel and mL is left. $\frac{738}{125}$ = 5.904 mL, reported as 5.9 mL; one 30 mL vial is enough.

> **DOES THE NUMBER MAKE SENSE?**
> 125 mg in each mL, so 738 mg is a little under 6 mL; 5.9 mL fits and is less than one 30 mL vial.

---

d

Her key

Answer: 14.77 mg/L

WHAT THE STEM GIVES

weight	162 lb	*not needed: C_{ss} is the target*
t½	3.1 hours	$k = \frac{0.693}{3.1} = 0.2235\ \text{hr}^{-1}$
C_{ss}	20 mcg/mL	the plateau, 20 mg/L
t	6 hour	the length of the infusion
age	58 years old	*not needed*
vial	30 mL	*not needed*
concentration	125 mg/mL	*not needed*
V_D	0.5 L/kg	*not needed: C_{ss} is the plateau*

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_p = C_{ss}(1 - e^{-kt}),\ \text{with}\ k = \frac{0.693}{3.1\ \text{hr}} = 0.2235\ \text{hr}^{-1}$$

Without a loading dose the level rises from zero toward C_{ss}; $(1 - e^{-kt})$ is the fraction of C_{ss} reached after t hours of infusion.

ALGEBRA

$$C6 = (20\ \text{mg/L})(1 - e^{-(0.2235)(6)}) = (20\ \text{mg/L})(0.7384) = 14.77\ \text{mg/L}$$

kt = 1.341 has no units, so the answer keeps the mg/L of C_{ss}. Six hours is just under two half-lives, so 73.84% of C_{ss} is reached.

> **DOES THE NUMBER MAKE SENSE?**

> Six hours is just under two half-lives of 3.1 hours, so the level is a little under three quarters of 20 mg/L: 14.77 mg/L fits.

---

e

Her key

Answer: 1.01 mg/L

WHAT THE STEM GIVES

t½	3.1 hours	$k = \frac{0.693}{3.1} = 0.2235\ hr^{-1}$
infusion	6-hour	ended at 14.77 mg/L
Cstop	14.77 mg/L	the starting point of the decline
t	12 hours	after the cessation: the exponent kt
weight	162 lb	*not needed*
age	58 years old	*not needed*
vial	30 mL	*not needed*
concentration	125 mg/mL	*not needed*
V_D	0.5 L/kg	*not needed: the decline needs only k*

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_p = Cstop\ e^{-kt}, \text{ with } k = \frac{0.693}{3.1\ hr} = 0.2235\ hr^{-1}$$

Once the infusion stops nothing enters, so the level falls first order from the end-of-infusion concentration; t counts from the stop.

ALGEBRA

$$C12 = (14.77\ mg/L)\ e^{-(0.2235)(12)} = (14.77\ mg/L)(0.06842) = 1.01\ mg/L$$

kt = 2.682 has no units; $e^{-2.682} = 0.06842$, so about 7% of the end-of-infusion level remains.

> **DOES THE NUMBER MAKE SENSE?**
>
> Twelve hours is nearly four half-lives of 3.1 hours, so a little more than one sixteenth of 14.77 mg/L remains, about 1 mg/L; 1.01 mg/L fits.

---

f

Her key

Answer: 6.2 hr

WHAT THE STEM GIVES

t½	3.1 hours	$k = 0.2235\ hr^{-1}$; two half-lives take the level to a quarter
C_{ss}	20 mcg/mL	the level when the infusion stops
target	5 mcg/mL	a quarter of 20
weight	162 lb	*not needed*
age	58 years old	*not needed*
vial	30 mL	*not needed*

concentration	125 mg/mL	*not needed*
V_D	0.5 L/kg	*not needed: the decline needs only k*

THE WORKING, ONE LINE AT A TIME

SETUP

With the loading dose the level is C_{ss}, 20 mg/L, when the infusion stops; then $C_p = C_{ss}\, e^{-kt}$

An appropriate loading dose holds the level at C_{ss} from the start, so the decline begins at 20 mg/L.

ALGEBRA

$$t = \frac{\ln(20 \div 5)}{0.2235\ \text{hr}^{-1}} = \frac{1.386}{0.2235\ \text{hr}^{-1}} = 6.2\ \text{hr}$$

The mg/L cancel inside the log, leaving a pure number; dividing by hr^{-1} gives hours. $k = \frac{0.693}{3.1}\ \text{hr} = 0.2235\ \text{hr}^{-1}$.

ALGEBRA

$\frac{20}{5} = 4$, two halvings: 2(3.1 hr) = 6.2 hr

Her key also counts half-lives: falling to a quarter takes two halvings of 3.1 hours.

> **DOES THE NUMBER MAKE SENSE?**
> From 20 to 5 mg/L is a fall to a quarter, two half-lives of 3.1 hours: 6.2 hr.